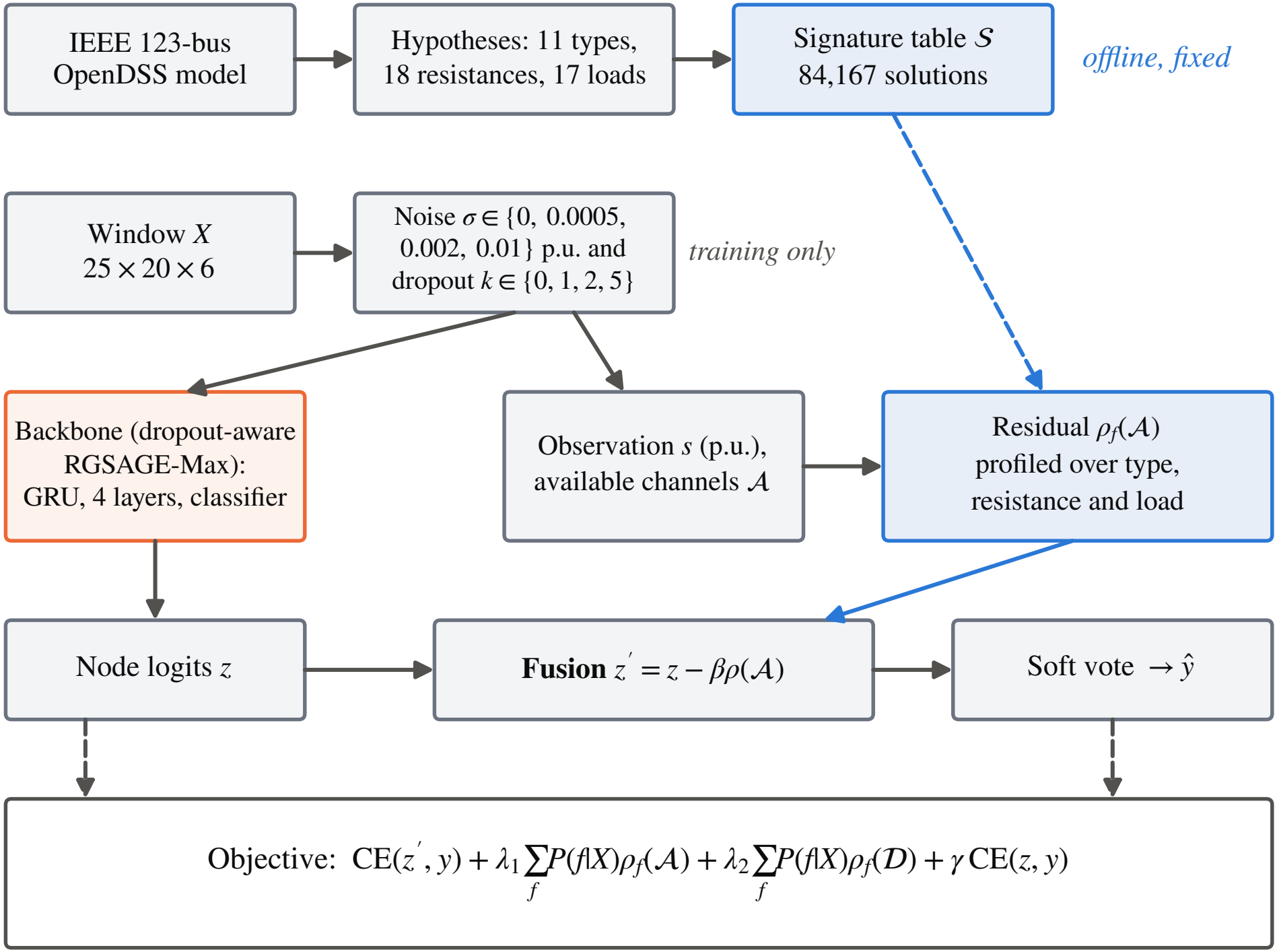
IEEE 123-bus
OpenDSS model
Hypotheses: 11 types,
18 resistances, 17 loads
Signature table $\mathcal{S}$
84,167 solutions
offline, fixed
Window X
$25 \times 20 \times 6$
Noise $\sigma \in \{0,\ 0.0005,$
$0.002,\ 0.01\}$ p.u. and
dropout $k \in \{0, 1, 2, 5\}$
training only
Backbone (dropout-aware
RGSAGE-Max):
GRU, 4 layers, classifier
Observation s (p.u.),
available channels $\mathcal{A}$
Residual $\rho_f(\mathcal{A})$
profiled over type,
resistance and load
Node logits z
Fusion $z' = z - \beta\rho(\mathcal{A})$
Soft vote $\rightarrow \hat{y}$
Objective: $\mathrm{CE}(z', y) + \lambda_1 \sum_f P(f|X)\rho_f(\mathcal{A}) + \lambda_2 \sum_f P(f|X)\rho_f(\mathcal{D}) + \gamma\,\mathrm{CE}(z, y)$